

Evaluation Protocol Choices Dominate Gene Regulatory Network Benchmarking Outcomes for Single-Cell Foundation Models

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ABSTRACT

Recent benchmarks report that single-cell foundation models such as scGPT and Geneformer recover gene regulatory networks (GRNs) from attention weights at performance competitive with established baselines. We show that this conclusion is governed primarily by three evaluation-protocol choices—gene-symbol mapping, the candidate edge set, and the gold standard—rather than by the models. We introduce the *universe-aware* protocol that scores every pair in $U = T \times (G \setminus \{\text{self}\})$ where T is a pinned human-TF list and G is the set of HGNC-approved expressed genes, and apply it to BEELINE hESC and hHep with cell-type ChIP-seq, non-specific ChIP-seq, and STRING gold standards. Across 279 (dataset, gold standard, candidate set, method) tuples, an ANOVA decomposition of $\log_{10}(\text{AUPR})$ assigns 80.1% of the variance to protocol-related factors and 0.27% to method identity. Restricting the candidate set to the literature-standard $T \times T_{\text{gold-targets}}$ inflates AUPR by 13–17 $\times$ without changing the underlying scores; AUPR tracks base rate at $\rho = 0.997$. Under universe-aware scoring against cell-type ChIP-seq, all six scGPT attention-aggregation variants have AUROC below 0.5 on hHep (95% DeLong CI strictly below 0.5, $p \ll 0.001$), while the same scGPT predictions reach AUROC 0.58 on hESC; the verdict flips by dataset and is invisible under aggregated single-number benchmarks. We release a versioned pipeline and a six-item reproducibility checklist. Code: <https://github.com/Biodyn-AI/grn-evaluation-bias>.

KEYWORDS

gene regulatory networks, single-cell foundation models, benchmarking, scGPT, Geneformer, reproducibility

1 UNIVERSE-AWARE EVALUATION

For a single-cell dataset D let $G(D)$ be the HGNC-approved expressed genes after BEELINE quality control under a SHA-pinned symbol dictionary, and let $T(D) = T_{\text{BEELINE}} \cap G(D)$ be the intersection of the 1,563-entry BEELINE human-TF list with $G(D)$. The *universe-aware* candidate set is

$$U(D) = \{(s, t) : s \in T(D), t \in G(D), s \neq t\}.$$

For BEELINE **hESC** (17,735 genes $\times$ 758 cells, [6]): $|G|=17,724$, $|T|=1,459$, $|U|=25,857,857$; **hHep** (11,515 $\times$ 425, [7]): $|G|=11,512$, $|T|=1,095$, $|U|=12,604,545$. All methods score *every* pair in $U(D)$. The TF list T_{BEELINE} is the human-TF compendium derived from [15] and shipped with BEELINE.

Methods compared. scGPT [1] (whole-human, 12 layers, 8 heads) and Geneformer-V1-10M [2] (6 layers, 4 heads) attention

Table 1: Universe-aware AUPR and AUROC, cell-type ChIP-seq. Foundation models: median (range) across six aggregations. †: 95% DeLong CI for AUROC lies entirely below 0.5.

Dataset	Method	AUPR	AUROC
hHep $b=0.018$ $ U =12.6\text{M}$	Spearman	0.020	0.526
	Geneformer (md.)	0.020	0.531 (.529-.534)
	Pearson	0.019	0.518
	random	0.018	0.501
	GRNBoost2	0.018	0.506
	scGPT (md.)	0.018	0.497[†] (.493-.499)
hESC $b=0.013$ $ U =25.9\text{M}$	Geneformer (md.)	0.020	0.638 (.635-.640)
	scGPT (md.)	0.017	0.578 (.577-.580)
	Pearson	0.015	0.550
	Spearman	0.015	0.544
	GRNBoost2	0.014	0.522
	random	0.013	0.500

with six aggregation variants; baselines Pearson, Spearman, GRNBoost2 [4, 5] (100 trees), and a seeded random scorer. AUROC significance uses DeLong’s analytical SE [13]; AUPR difference uses 2,000-resample paired bootstrap.

2 RESULTS

Protocol dominates model. A Type-II ANOVA on $\log_{10}(\text{AUPR})$ across 279 (dataset, gold standard, candidate set, method) tuples assigns 57.5% of variance to gold standard and 22.6% to candidate set (cumulative 80.1% protocol), versus 0.27% to method identity—a $\sim 300\times$ gap. Restricting the candidate set from $T \times G$ to $T \times T_{\text{gold-targets}}$ inflates median AUPR by 12.6 $\times$ on hESC and 17.0 $\times$ on hHep with the same scores; inflation is uniform across methods including the random baseline. Across all tuples, Pearson correlation between AUPR and base rate is $\rho = 0.997$.

Foundation-model verdict flips by dataset. Table 1 shows universe-aware AUPR and AUROC against cell-type-specific ChIP-seq. On hHep, all six scGPT aggregation variants have AUROC ≤ 0.499 with 95% analytical CI upper bound below 0.5 ($\sigma_{\text{AUROC}} \leq 6 \times 10^{-4}$, $p \ll 0.001$ vs $H_0 : \text{AUROC} = 0.5$), i.e., significantly *below random*; on hESC the same model reaches AUROC ~ 0.58 . Geneformer is moderate on hHep (AUROC 0.53) and strong on hESC (AUROC 0.64). On the broader STRING gold standard, no method is below random. The dataset $\times$ gold-standard sensitivity is invisible in aggregated benchmarks that report a single AUPR number, and the aggregation flatters the model because it averages “below-random on hHep” with “above-random on hESC”.

Aggregation choice does not rescue the result. All six aggregations fall within a tight range on universe-aware AUPR (scGPT hHep: 0.0183–0.0187; Geneformer hHep: 0.0195–0.0199); the “max-heads, mean-layers” variant cited in prior reports is statistically indistinguishable from the mean-by-mean default. The full Type-II decomposition is Table 2.

Table 2: Type-II sum of squares of $\log_{10}(\text{AUPR})$ across 279 (dataset, gold standard, candidate set, method) tuples. Protocol factors account for 80.1%; method identity for 0.27%.

Factor	SS	% total
Gold standard	42.88	57.5%
Candidate set	16.89	22.6%
Dataset (hESC vs hHep)	0.42	0.6%
Method family	0.20	0.27%
Residual	14.13	18.9%

Table 3: Universe-aware AUROC across three gold standards, median across attention-aggregation variants. The same model gives *opposite verdicts* (below vs above random, $\dagger = 95\%$ DeLong CI below 0.5) depending on the gold standard.

Tag	Method	cell-type ChIP-seq	non-specific ChIP-seq	STRING
hHep	scGPT	0.497 $\dagger$	0.476 $\dagger$	0.579
	Geneformer	0.531	0.499	0.595
	GRNBoost2	0.506	0.501	0.543
	Pearson	0.518	0.508	0.555
	Spearman	0.526	0.519	0.590
hESC	scGPT	0.578	0.523	0.604
	Geneformer	0.638	0.550	0.665
	GRNBoost2	0.522	0.501	0.544
	Pearson	0.550	0.517	0.585
	Spearman	0.544	0.513	0.575

Symbol mapping shifts coverage. Under the pinned HGNC dictionary, 89–93% of raw symbols resolve as approved, 7–10% via previous/alias resolution, and $\sim 1\%$ are unmapped. Without HGNC mapping, edge coverage between predicted edges and the cell-type ChIP-seq gold is effectively zero; with mapping it exceeds 99%. Symbol-mapping choices are therefore necessary upstream of every numeric claim in this and prior reports; we ship the dictionary as a SHA-checked file.

Gold-standard sensitivity is the third axis of protocol bias. The same scGPT predictions move from *below random* on hHep cell-type ChIP-seq to clearly above random on STRING [16] and on hESC cell-type ChIP-seq (Table 3). Geneformer similarly shifts by 0.07–0.11 AUROC across gold standards on the same dataset. Alternative regulatory references such as DoRothEA [8] and TRRUST [9] use yet different inclusion criteria and would add further variability we do not estimate here. Reports that cite a single “AUPR against ChIP-seq” number for a foundation model are not falsifiable without disclosing the gold-standard identity, the candidate set, and the universe size.

Implications for benchmark design. A defensible foundation-model GRN claim must specify the pinned symbol dictionary, the candidate universe U (cardinality and base rate), and the gold standard with its source, cell-type, and version. The literature today routinely specifies only the last of these three, often loosely; our released pipeline emits all three by construction.

Top-of-ranking precision is even harsher than AUROC. At the top of the universe-aware ranking on hHep cell-type ChIP-seq (base rate 0.018), precision@50 is zero for 5 of 6 scGPT variants and for Pearson and Spearman; precision@1000 is 0.009–0.017 for scGPT versus 0.026 for random. Geneformer fares better (~ 1.0 – $1.6\times$ random). For both foundation models on hHep, the top of the ranked attention edge list is indistinguishable from—or worse than—a random shuffle.

3 REPRODUCIBILITY CHECKLIST

(1) pin and ship a SHA-checked gene-symbol dictionary (here: HGNC [14] dump SHA-256 b9482a52. . . , 50,094 rows); (2) define and report $|G|$, $|T|$, $|U|$, and base rate for every metric; (3) report AUPR with its candidate-set definition—never as a single number—and report AUROC alongside; (4) declare “below random” only when an analytical or bootstrap-derived 95% CI for the relevant statistic excludes the null; (5) include both a correlation baseline (Pearson/Spearman) and a tree-based baseline (GRNBoost2/GENIE3) under the same universe; (6) release executable pipelines with frozen dependency versions, seeded permutation/bootstrap, and SHA-checked data archives (here: BEELINE Zenodo record 3701939, SHA 1fc84949. . . for inputs, d54fc6ed. . . for networks).

Limitations. Two human BEELINE datasets; mouse benchmarks and DREAM5 left for future work. scGPT caps per-cell sequence at 1,200 tokens; cell-aggregated attention covers all expressed genes. Results may differ for Geneformer-V2 (316M parameters) we did not run. BEELINE expression is log-normalized; we invert via $\text{round}(\expm1(\cdot))$ for tokenizer-based models.

Our analysis does *not* preclude the possibility that foundation-model representations contain regulatory signal extractable by more sophisticated decoders; it establishes that raw attention aggregation does not suffice and that current benchmark claims require re-examination under controlled evaluation protocols. The point that protocol choices propagate into reported model rankings echoes earlier critiques of trajectory [10] and DREAM network [11] inference; we quantify the effect for transformer attention against ChIP-seq [3, 12] and protein-association gold standards under a fully pinned pipeline.

Code and data. Full MIT-licensed pipeline (BEELINE preprocessing, scGPT and Geneformer attention extraction, baselines, universe-aware evaluation, DeLong AUROC CI with paired AUPR bootstrap, Type-II ANOVA, paper-number trace) at <https://github.com/Biodyn-AI/grn-evaluation-bias>, with the pinned HGNC dictionary, BEELINE SHA checks, and a conda lockfile. All numbers above are emitted verbatim by the pipeline.

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